

M531X

MQTT User Guide

NB-IoT Series

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Services & Supports

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About This Document

M531X NB-IoT series of products contained herein refer to M5310-A, M5311, M5312, M5313 and N10SG modules.

Revision History

Version	Date	Author	Description
V1.0.0	2018-06-11	Lin Ziwei	Initial version.
V1.1.0	2018-08-20	Lin Ziwei	Added "Statement Page". Added \r\n at the end of +MQTTPUBLISH.
V1.2.0	2018-09-12	Lin Ziwei	Revised the error in the parameters notified by URC.
V1.3.0	2018-09-19	Lin Ziwei	- Added the description of M531X series; - Modified the wrong description of the value returned by +MQTTSUB; - Corrected the wrong description of one parameter of +MQTTPUB.
V1.4.0	2019-02-17	Li Yanlun	- +MQTTPU default msg_len changed to non-default; - Modified MQTTSTAT description; - Encrypt revised as not supported temporarily.
V1.4.1	2021-04-02	Li Yanlun	Modified the parameter info in Section 1.5

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1 M531X Series MQTT AT Command

1.1 AT+MQTTCFG

This command configures MQTT connection parameter.

AT+MQTTCFG	
Syntax	
Set command AT+MQTTCFG=<host>,<port>,<id>,<keepAlive>,<user>,<passwd>,<clean>[,<encrypt>]	Response: OK
Read command AT+MQTTCFG?	Response: AT+MQTTCFG:<host>,<port>,<id>,<keepAlive>,<user>,"*****",<clean>,<encrypt>
Parameter Description	
<host>	
IP or domain name address of MQTT Server.	
<port>	
Port of MQTT Server.	
<id>	
Client ID.	
<keepalive>	
Keepalive time.	
<user>	
User name.	
<passwd>	
Password.	
<clean> Whether to delete session.	
0	Not deleted
1	Delete
<encrypt>	
Whether to enable TLS encryption (1 is not supported yet).	
Example	
AT+MQTTCFG="183.230.40.39",6002,"4069959",60,"75829","lIOu0oFUg1guk20ornTK1uzAcnM=",1 OK AT+MQTTCFG? +MQTTCFG:183.230.40.39,6002,4069959,60,75829,***,1 OK	



If the keepAlive time is less than ACK response detection time (set through AT+MQTTTO), PINGREQ may not detect network disconnection.



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1.2 AT+MQTTOPEN

This command connects MQTT server.

AT+MQTTOPEN	
Syntax	
Set command AT+MQTTOPEN=<usrFlag>,<pwdFlag>,<willFlag> >,<willRetain>,<willQos>,<will-topic>,<will-mesg>	Response: OK
Parameter Description	
<usrFlag>	
This flag bit represents that a Client needs a user name.	
<pwdFlag>	
This flag bit represents that a Client needs a password.	
<willFlag>	
If this bit is set to 1, a Will Message will be published when an I/O error occurs while the Server communicates with the Client or the Client fails to communicate within the Keep Alive time. When the Server receives DISCONNECT message, it will not trigger the Server to send a Will message.	
<willRetain>	
This bit specifies if the Will Message published by the Server is to be retained if Client connection is accidentally interrupted.	
<willQos>	
This bit specifies the QoS level to be used when publishing the Will Message.	
<will-topic>	
The topic name of Will message.	
<will-mesg>	
Will message.	
Example	
AT+MQTTOPEN=1,1,0,0,0,"","	//User name and password are required (set in the CFG configuration command). No will setting.
OK	
+MQTTOPEN: OK	



For relevant command parameters, please refer to MQTTv3.1 protocol. Server connection failure (non password validation error)/ACK timeout/data sending failure/PING timeout will result in reconnection of Client. Client reconnection stops after several attempt failures and reports the +MQTTDISC message. Then, it is up to the customer to reconnect or not.

1.3 AT+MQTTSTAT

This command inquires the connection status of MQTT.

AT+MQTTSTAT	
Syntax	
Query command AT+MQTTSTAT?	Response: +MQTTSTAT:<stat> OK
Parameter Description	
<stat> Present connection status	
0	Client parameter not initialized.
1	Client parameter initialized.
2	Disconnected with the Server.
3	Sending Connect packet, waiting for receiving Server ack.
4	Reconnecting to the Server.
5	Connected to the Server.
6	Establishing TCP connection.
7	TCP connection established.



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1.4 AT+MQTTSUB

This command subscribes messages.

AT+MQTTSUB	
Syntax	
Set command AT+MQTTSUB=<topic>,<qos>[,<index>]	Response: OK If the Server subscribes ACK information, then URC +MQTTSUBACK: <packId>,<code>,<topic> is received.
Read command AT+MQTTSUB?	Response: +MQTTSUB: <subscribe list>
Parameter Description	
<topic>	
Topic of subscribed message.	
<qos>	
The level of quality for delivery of the subscribed message, 0 - 2.	
<packId>	
Data packet ID.	
<Index>	
Operation Index when the message published by topic is received. At present, it only supports 0, meaning the published message is output through AT port.	
<code> Server feedback code	
0x00	QoS0 subscribe success
0x01	QoS1 subscribe success
0x02	QoS2 subscribe success
0x80	Failure
<subscribe list>	
List of subscribed messages.	
Example	
AT+MQTTSUB="dev/gsm/u",2 OK +MQTTSUBACK:2,2,dev/gsm/u AT+MQTTSUB? +MQTTSUB: dev/gsm/u OK	



The subscribed message will be saved in the subscribe list. Only one subscription can be added at a time currently.

1.5 AT+MQTTPUB

This command publishes the message.

AT+MQTTPUB	
Syntax	
Set command AT+MQTTPUB=<topic>,<qos>,<retain> ,<dup>,<msg_len>,<message>	Response: OK
Parameter Description	
<topic>	
Topic of subscribed message.	
<qos>	
The level of QoS for delivery of the subscribed message, 0 to 2 are supported.	
<retain>	
This parameter indicates if the server retains this message and sends the latest subscribed message to the new subscriber. It can be set to 0 and 1.	
<dup>	
Duplicate flag, 0-1.	
<msg_len> Type or length of the input message	
0	Character string format.
Non-zero	Divide the input length of hexadecimal string by 2.
<message>	
Contents of message.	
Example	
AT+MQTTPUB="dev/gsm/u",2,1,0,0,"update message"	
OK	
+MQTTPUBREC:10,0	
+MQTTPUBCOMP:10,0	
AT+MQTTPUB="dev/gsm/u",2,1,0,2,"1890"	
OK	
+MQTTPUBREC:11,0	
+MQTTPUBCOMP:11,0	



DUP flag will be set to 0 internally.

1.6 AT+MQTTUNSUB

This command unsubscribes.

AT+MQTTUNSUB	
Syntax	
Set command AT+MQTTUNSUB=<topic>	Response: OK When receive ACK of unsubscribe message from server, +MQTTUNSUBACK:<packId>,<topic> will be returned.
Parameter Description	
<topic>	
Topic of subscribed message.	
<packId>	
Data packet ID.	
Example	
AT+MQTTUNSUB="dev/gsm/u" OK +MQTTUNSUBACK: 8,"dev/gsm/u"	



Only one subscription can be canceled at a time currently.

1.7 AT+MQTTDISC

This command actively disconnects and sends the DISCONNECT packet.

AT+MQTTDISC	
Syntax	
Execute command AT+MQTTDISC	Response: OK When the connection is disconnected, +MQTTDISC: OK is returned, indicating the connection is already disconnected. If this command is used when the current connection is not established, ERROR is returned.
Example	
AT+MQTTDISC OK +MQTTDISC: OK	

1.8 AT+MQTTDEL

This command deletes the instances of MQTT Client and releases resources.

AT+MQTTDEL	
Syntax	
Execute command AT+MQTTDEL	Response: OK
Example	
AT+MQTTDEL OK	

1.9 AT+MQTTTO

This command sets ACK timeout period that should be less than the keepalive time.

AT+MQTTTO	
Syntax	
Set command AT+MQTTTO=<timeout>	Response: OK
Automatic report	
+MQTTTO:<type>	
Parameter Description	
<timeout>	
Timeout period. If the Client hasn't received the Server ACK within the timeout period, +MQTTTO will be reported. Default 10s.	
<type> Timeout type.	
1	Connection timeout
2	Publish response timeout
3	Subscribe response timeout
4	Unsubscribe response timeout
5	Ping timeout
6	Unknown type

1.10 AT+MQTTPING

This command sets the ping packet ACK display. Reconnection will be triggered when no Server ACK is detected.

AT+MQTTPING	
Syntax	
Set command AT+MQTTPING=<rspEcho>	Response: OK
Parameter Description	
<rspEcho>	
This parameter indicates whether to echo +MQTTPINGRSP message. Default 0.	
Example	
+MQTTPINGRSP: OK	//If ping packet response result is received when echo function is enabled.
+MQTTPINGRSP: OK	

1.11 MQTT URC Report

MQTT relevant commands, URC automatic report.

MQTT URC Report	
Syntax	
	+MQTTPUBLISH:<dup>,<qos>,<retained>,<packId>,<topic>,<msg_len>,<message> If the topic of the subscribed message is pushed, this message will be reported automatically. +MQTTSUBACK: <packId>,<code>,<topic> code response values are as follows: 0x00 Maximum QoS0 subscribe Success 0x01 Maximum QoS1 subscribe Success 0x02 Maximum QoS2 subscribe Success 0x80 Failure Server ACK message is reported of subscribing message. +MQTTPUBACK:<packId>,<dup> Message publish response ACK reported under QoS1 mode. PackId refers to the packet number. +MQTTPUBREC:<packId>,<dup> Publish response 2 under QoS2 mode. +MQTTPUBREL:<packId>,<dup> Publish response 3 under QoS2 mode. +MQTTPUBCOMP:<packId>,<dup> Publish response 4 (last) under QoS2 mode. +MQTTPINGRSP:OK Client ping packet ACK. Set whether to report it automatically through AT+MQTTPING command. +MQTTUNSUBACK:<packId>,<topic> Actively report of unsubscribing ACK. +MQTTTO: <type> Timeout report. +MQTTDISC: Report actively when the connection is disconnected.
Example	
<pre>AT+MQTTSUB="dev/gsm/u",2 OK +MQTTSUBACK:8,2,dev/gsm/u AT+MQTTPUB="dev/gsm/u",2,1,0,"update message" //Send data to dev/gsm/u topic. OK +MQTTPUBREC: 0,14 +MQTTPUBCOMP: 0,14 +MQTTPUBLISH: 0,2,0,9,dev/gsm/u,14,update message</pre>	

2 MQTT Application Routine

Unless otherwise noted, M5310-A module is taken as the example in the following AT examples.

2.1 MQTT Client Parameter Configuration

AT+MQTTCFG="183.230.40.39",6002,"4069959",10,"75829","IIOu0oFUg1guk20ornTK1uzAcnM=",1

2.2 Connect -Subscribe - Push - Unsubscribe

AT+MQTTOPEN=1,1,1,1,1,"mywill","001bye"	//Send connection establishment request. Will topic name:
OK	mywill; will message content: 001bye.
+MQTTOPEN:OK	//Connection succeeded.
AT+MQTTSUB="pyr",1	//Subscribe topic pyr.
+MQTTSUBACK:1,1,pyr	//Subscribe success response.
AT+MQTTPUB="dev/gsm/u",1,1,0,0,"abc"	//Publish data. Data content is character string abc.
+MQTTPUBACK:1,0	//Publish data response, published successfully.
AT+MQTTPUB="pyr",1,0,0,3,"7E7A7A"	//Publish data to the subscribed topic "pyr", input hexadecimal mode. At the moment, the Client will receive the publish data sent by the Server.
	//Publish data response.
+MQTTPUBACK:3,0	//Receive the published data of the subscribed topic "pyr" at the same time.
+MQTTPUBLISH:0,1,0,1,pyr,3,~zz	
AT+MQTTPUB="pyr",1,0,0,0,"abcdef"	//Publish data to the subscribed topic and enter character string data.
+MQTTPUBACK:4,0	//Receive the publish response.
+MQTTPUBLISH:0,1,0,1,pyr,6,abcdef	//Receive and send the published message of topic pyr.
AT+MQTTUNSUB="pyr"	//Unsubscribe.
+MQTTUNSUBACK:5,pyr	//Unsubscribe response.
AT+MQTTPUB="pyr",1,0,0,0,"abcdef"	//Publish data to pyr again.
+MQTTPUBACK:6,0	//Publish response. At this time, pyr published message will not be received.

2.3 Disconnect - Delete Parameters

AT+MQTTDISC	//Disconnect.
AT+MQTTDEL	//After releasing resources, +MQTTCFG can be used again for parameter configuration.

2.4 Description about Disconnection

When applying the MQTT function of the module, the ping packet will be sent at the KeepAlive interval configured in +MQTTCFG to test the connection status when the module isn't processing data (that is to say, no data packet being sent from the Server.) When the ping packet is not reached continuously for three times (interval: KeepAlive), the reconnection mechanism will be triggered to reconnect at regular time. After three attempt failures, reconnection will be disabled and subsequent strategy should be formulated according to the application scene, e.g., restarting the module or resending +MQTTOPEN to initiate connection request after an interval. Active disconnection through AT+MQTTDISC command will not trigger reconnection.

+MQTTPINGRESP:OK	//Ping response received.
+MQTTPINGRESP:OK	//Ping response received.
+MQTTTO:5	//First ping timeout.
+MQTTTO:5	//Second ping timeout.
+MQTTTO:5	//Third ping timeout.
+MQTTREC:0	//First reconnection.
+MQTTREC:1	//Second reconnection.
+MQTTREC:2	//Third reconnection.
+MQTTDISC:OK	//Reconnection failed. Then, disconnection prompt.

